COURSE GUIDE & SYLLABUS · KG–GRADE 2 · AGES 6–7

Scratch Junior (K–2) Advanced: Game Builders

Tap starts, forever loops, scores, messages and bumps — build your first real games.

The third ScratchJr course for ages 6–7. Children learn how games work: tap-to-start interaction, forever loops, keeping score, planning paths through mazes, messages between characters, bump (collision) rules, designing fair levels, and testing and fixing games. Projects: Tap Game, Gem Collector, Relay Race and a capstone My First Game. Every lesson has a short video, a build-along for the ScratchJr app, picture-block puzzles that check themselves, an active warm-up, a worksheet and a teacher plan.

What learners will be able to do

- Start scripts with a tap as well as the green flag
- Choose between counted and forever loops
- Keep score by adding 1 for each thing collected
- Plan paths through mazes and find shorter ways
- Use messages so characters take turns
- Make game rules with bumps (collisions)
- Design fair levels that get harder
- Test games and fix bugs in rules and scores
- Design, build and present an original game

How every lesson works

Watch	A short animated explainer video with captions (and an optional read-aloud).
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.
Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Tap to Start** 👆 Start a script by tapping a character, and choose between the green flag and tap starts.	Events (tap to start)	A character that reacts
2	**Repeat Forever** ♾️ Tell the difference between repeating a set number of times and forever, and choose the right loop for a game.	Forever loops	Forever fish
3	**Project: Tap Game** Plan and build a simple tap game where characters react in fun ways when the player taps them.	Project: interaction	Project: Tap Game
4	**Keep the Score** ⭐ Add 1 to a score each time a gem is collected, and check the score at the end.	Counting (variables)	Score keeper
5	**Mazes and Smart Paths** Plan a path through a maze, avoiding walls, and find a short way.	Algorithms (planning paths)	Solve a maze
6	**Project: Gem Collector Game** Plan and build a game where a character collects treasure, the score goes up and the game says when you win.	Project: score and paths	Project: Gem Collector Game
7	**Send a Message** ✉️ Explain how one character can send a message that starts another character's script, and plan a relay.	Messages (communication between characters)	A relay race in ScratchJr
8	**Bump! Characters That Touch** Make something happen when two characters touch (bump), and use bumps in game rules.	Collisions (start on bump)	Bump to collect
9	**Project: Relay Race** Plan and build a relay race where runners pass a message to start the next runner, ending with a winner.	Project: messages	Project: Relay Race
10	**Design a Game Level** Design a fair level with a path, treasure and a goal, then solve it with as few blocks as you can.	Design and efficiency	Design and solve
11	**Test and Fix Your Game** Test a game like a player, find bugs in rules and scores, and fix them.	Testing and debugging	Fix the games
12	**Capstone: My First Game** Design, build, test and present a ScratchJr game with	Capstone project	Project: My First Game

	a start, player interaction, a bump rule and a win message.		

Modules

Module 1 · Game starts
Tap to start and forever loops. Project: Tap Game.

Big question: How do players make games happen?

Module 2 · Score and paths
Keeping score and solving mazes. Project: Gem Collector Game.

Big question: How do games count and challenge us?

Module 3 · Messages and bumps
Messages between characters and bump rules. Project: Relay Race.

Big question: How do characters work together?

Module 4 · Make a real game
Level design, testing and a capstone game.

Big question: How do game designers make games fun and fair?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Tablet or computer with a modern browser
- Optional: the free ScratchJr app (iPad, Android tablet or Chromebook) for the projects
- Paper and crayons for game plans

Before you start

- Scratch Junior (K–2) Intermediate: Story Makers, or confident use of ScratchJr moves, looks, repeat and pages

Safety and privacy

Children use only first names and never share photos of themselves or others outside the class. ScratchJr projects stay on the device. Young learners use the AI tutor only with an adult present

and guardian consent.

What comes next

After this course: **Scratch (Gr 3–5) Beginner**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.